DEPARTMENT OF WORKS & HIGHWAYS

MATERIAL SCIENCE & TECHNOLOGY BRANCH

Materials Testing Laboratory

☑ Port Moresby ☐ Mt. Hagen, ☐ Lae ☐ Annex Lab.

Date:

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PARTICLE DENSITY AND WATER ABSORPTION OF FINE / COARSE AGGREGATE - AS1141.5 & 6.1

Client : _____ Asset No.: _____

Project: _____ Material Source : _____

Location: _____ Pit No.: _____

Material Description: _____ Depth: _____

FINE AGGREGATE

Flask No.			1	2	Temperature of water used = _____
M1	Mass of total test portion in oven - dry condition	g			
M2	Mass of total test portion in saturated surface dry	g			
M3	Mass of flask + test portion (SSD) + water	g			
M4	Mass of flask + water	g			
ρ_a	Apparent Particle Density of test portion ρ_a = (M1 x ρ water) / M4 + M1 - M3	t/m3			Ave. ρa t/m3
ρ_{bd}	Particle Density on dry basis ρ_{bd} = (M1 x ρ water) / M4 + M2 - M3	t/m3			Ave. ρbd t/m3
ρ_{bs}	Particle Density on SSD basis ρ_{bs} = (M2 x ρ water) / (M4 + M2 - M3	t/m3			Ave. ρbs t/m3
WA	**Water Absorption = (M2 - M1) x 100 / M1**	**%**			Ave. **Wa** = %

COARSE AGGREGATE

Number of test portion			1	2	Temperature of water used = _____
W1	Mass of soaked aggregate + basket in water	g			
W2	Mass of wire basket in water	g			
M2	Mass of saturated surface - dry aggregate in air	g			
M1	Mass of oven - dry aggregate in air	kg			
ρ_A	Apparent Particle Density ρ_A = (M1 x ρ water) / M1 - (W1 - W2)	t/m3			Ave. ρA t/m3
ρ_D	Particle Density on dry basis ρ_D = (M1 x ρ water) / M2 - (W1 - W2)	t/m3			Ave. ρD t/m3
ρ_S	Particle Density on a SSD basis ρ_S = (M2 x ρ water) / M2 - (W1 - W2)	t/m3			Ave. ρS t/m3
WA	**Water Absorption = (M2 - M1) x 100 / M1**	**%**			Ave. **WA** = %

COMBINED TEST FRACTIONS CALCULATIONS

ρw(n) = Weighted average density (app, dry or ssd) = (100xρcoarse x ρfine) / (%coarse x ρfine + %fine x ρcoarse)

= t/m3

WAA = weighted average water absorption in % = (Absorption coarse x % coarse + Absorption fine x % fine) / 100

= %

"NOTE: ρ = DENSITY"

Tested by: _____ Checked by: _____

Date: _____ Date: _____

Authorised by: AS(Material Science & Technology Branch)

FORM No. A-5

Revised Worksheet: 22/06/2022